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C1 Pathway Designer

Multi-Substrate C1 Pathway Platform

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C1 Pathway Design

Designed for CO₂, methanol, formaldehyde, and formate substrates, the platform systematically screens C1 assimilation pathways to target products through candidate-reaction combination enumeration and pFBA feasibility analysis.

[Start Calculation](#)[View Results](#)

Workflow

01

Select Substrate and Product

Choose a C1 substrate and the target metabolite to produce.

02

Set Combination Size

Specify the number of candidate reactions opened simultaneously.

03

Enumerate Combinations

Generate combinations from the substrate-specific candidate pool.

04

pFBA Feasibility Screening

Determine whether each combination supports target production.

05

Analyze and Visualize

Output fluxes, thermodynamic plots, MDF plots, and pathway networks.

Platform Capabilities

Unified Multi-Substrate Framework

Supports CO₂, methanol, formaldehyde, and formate as C1 substrates.

Reaction Combination Search

Enumerates substrate-specific candidate pools and compares pathway feasibility.

Flux and Thermodynamics

Provides pFBA flux tables, standard reaction Gibbs energy plots, and MDF plots.

Visualization and Annotation

Displays pathway networks with reaction equations, fluxes, and EC numbers.

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Supporting C1 pathway enumeration, pFBA screening, thermodynamic analysis, and network visualization.

B Task submission

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Submit a Pathway Search Task

Select a C1 substrate, target product, candidate-reaction combination size, and maximum pathway length. The system enumerates the complete substrate-specific candidate pool and uses pFBA to identify all qualifying feasible pathways.

Task Parameters

Substrate

Formaldehyde

Product

3-Phospho-D-glycerate (C00197)

C1 Reactions per Combination

1

Maximum Pathway Reactions

20

Email Notification

Optional, e.g. user@example.com

[Submit Task](#)[Reset Parameters](#)

Parameter Guide

Substrate / Product

Four C1 substrates are currently supported.

C1 Reactions per Combination

Simultaneously opens k candidate reactions and tests whether the combination supports production of the target product.

Maximum Pathway Reactions

Retains all feasible pathways whose length is less than or equal to this main-pathway length.

Email Notification

If SMTP is configured, a notification is sent when the task finishes; otherwise, the address is only recorded.

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C Calculation results

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Calculation Results

Browse and manage calculation tasks, expand details to inspect candidate pathways, and open pathway visualizations.

All 48

Running 0

Completed 48

Search by job ID, substrate, or product

Refresh

job id	substrate	product	C1 reactions per combination	candidate pool size	maximum pathways	status	timestamp	action
dc58e4f578ba	Formaldehyde	3-Phospho-D-glycerate (C00197)	1	13	1	Completed	2026-07-16T11:08:42	detail delete
Pathway Results (1) Collapse Details								
Pathway ID	Substrate	Product	Reaction Count	Delta G	MDF	action		
path_0001	Formaldehyde	3-Phospho-D-glycerate (C00197)	6	-102.76 kJ/mol	19.19 kJ/mol	Visualize	Download	
06a02bad4d0f	Formaldehyde	Pyruvate (C00022)	1	13	1	Completed	2026-07-16T11:01:25	detail delete
ea11d6d99f69	Formaldehyde	Pyruvate (C00022)	1	1	1	Completed	2026-07-16T10:54:29	detail delete
f5d08d8f3f0c	Formaldehyde	Pyruvate (C00022)	1	13	1	Completed	2026-07-16T10:58:41	detail delete
bd181027e032	Formaldehyde	3-Phospho-D-glycerate (C00197)	1	13	1	Completed	2026-07-16T10:33:39	detail delete
07d95166d408	CO2	Acetyl-CoA (C00024)	1	1	1	Completed	2026-06-30T13:40:14	detail delete
0c22d981c53e	CO2	Acetyl-CoA (C00024)	1	16	0	Completed	2026-06-30T13:40:14	detail delete
3d66d773e3e	CO2	Acetyl-CoA (C00024)	1	1	0	Completed	2026-06-30T13:40:14	detail delete
42696fa469c6	CO2	Acetyl-CoA (C00024)	1	16	1	Completed	2026-06-30T13:40:14	detail delete
aaa2781a1d27	CO2	Acetyl-CoA (C00024)	1	1	1	Completed	2026-06-30T13:40:14	detail delete

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D Thermodynamics and MDF

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Pathway Visualization

Select a pathway on the Results page to view thermodynamic plots, the MDF plot, the flux network, and reaction details.

Pathway ID	Substrate	Product	Reactions	Delta G	MDF
path_0001	Formaldehyde	3-Phospho-D-glycerate (C00197)	6	-102.76 kJ/mol	19.19 kJ/mol

Thermodynamics and MDF

Standard Gibbs Energy

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MDF

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E Pathway network

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Pathway Network

[Download Pathway Table](#)

[Save JSON](#)[Show Reaction Nodes](#)[Download SVG](#)

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F Reaction details

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Reaction Details

6 rows

Reaction ID	Equation
FL5	3.0 FORMALDEHYDE_c -> DIHYDROXYACETONE_c
PHOSPHOGLYCERATE-KINASE-GTP-RXN	G3P_c + GTP_c <=> DPG_c + GDP_c
GAPDHPHOSPH-RXN	GAP_c + NAD_c + P_i_c <=> DPG_c + NADH_c + PROTON_c
RXN-14117	GTP_c + PYRUVATE_c -> GDP_c + PHOSPHO_4S_ENOL_4S_PYRUVATE_c + PROTON_c
TRIOSEPHOSMERIZATION-RXN	GAP_c <=> DIHYDROXY_4S_ACETONE_4S_PHOSPHATE_c
2.7.1.125-RXN	DIHYDROXYACETONE_c + PHOSPHO_4S_ENOL_4S_PYRUVATE_c -> DIHYDROXY_4S_ACETONE_4S_PHOSPHATE_c + PYRUVATE_c

Reaction Details

Reaction ID	EC number	Flux	Standard ΔG
PHOSPHOGLYCERATE-KINASE-GTP-RXN	2.7.2.10	-3.33333	-22.03 kJ/mol

Equation
G3P_c + GTP_c <=> DPG_c + GDP_c

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